parameter updates, while the full gradient method uses all available data.

6406532878924. ✔ The semi-gradient method updates the parameters of the value function based on a mixture of TD error and the gradient of the value function, while the full gradient method updates parameters using only the gradient.

6406532878925. ✖ The semi-gradient method is more computationally efficient than the full gradient method, requiring fewer iterations to converge to the optimal solution.

6406532878926. ✖ The semi-gradient method guarantees convergence to the optimal policy for any choice of learning rate, while the full gradient method may diverge with certain learning rates.

Industry 4.0

Section Id :	64065361123
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	8
Number of Questions to be attempted :	8
Section Marks :	20
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653127960
Question Shuffling Allowed :	No

Question Number : 19 Question Id : 640653856775 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532878935. ✔ YES

6406532878936. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653127961
Question Shuffling Allowed :	Yes

Question Number : 20 Question Id : 640653856776 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

In a EOQ calculation, the estimated and actual parameter values are given in Table 1. What is the error in the EOQ calulation {error= ($Q_{estimated}$ - Q_{actual})/ Q_{actual} }. (round your answer to two decimal without the percentage sign e,g, enter 10.12 if your answer is 10.12 %, enter -10.12 if your answer is -10.12%)

Table 1

Parameter	Estimate	Actual
Annual demand	1000 units	2000 units
Holding cost per unit per year	$10	$20
Ordering cost per order	$50	$25

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

40.00 to 42.00

Sub-Section Number :	3
Sub-Section Id :	640653127962
Question Shuffling Allowed :	No

Question Id : 640653856777 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (21 to 22)

Question Label : Comprehension

Suppose you need to accept reservation for a flight. There is total 100 seats in the aircraft. You have two classes of passengers - economy and business. One economy seat gets you 2,500 Rs of revenue while a business seat gets 7,500 Rs. From your experience, you know the demand for business travellers follows a uniform distribution in [10,20]. Then answer the given subquestions:

Sub questions

Question Number : 21 Question Id : 640653856778 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

What is the Booking limit?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

17

Question Number : 22 Question Id : 640653856779 Question Type : SA

Correct Marks : 1.5

Question Label : Short Answer Question

What is the protection level?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

83

Sub-Section Number :	4
Sub-Section Id :	640653127963
Question Shuffling Allowed :	No

Question Id : 640653856780 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (23 to 26)

Question Label : Comprehension

Ramu need to visit City-7. Currently he is stationed at City-1. He has a few alternative routes denoted by the arrows (arcs). There are other cities viz. 2, 3, 4, 5, 6 which falls on various routes between City-1 and City-7. The weights on the arcs represents the distances between respective nodes. However, there is no connection between City-2 and City-6. This scenario is represented in Fig-1.

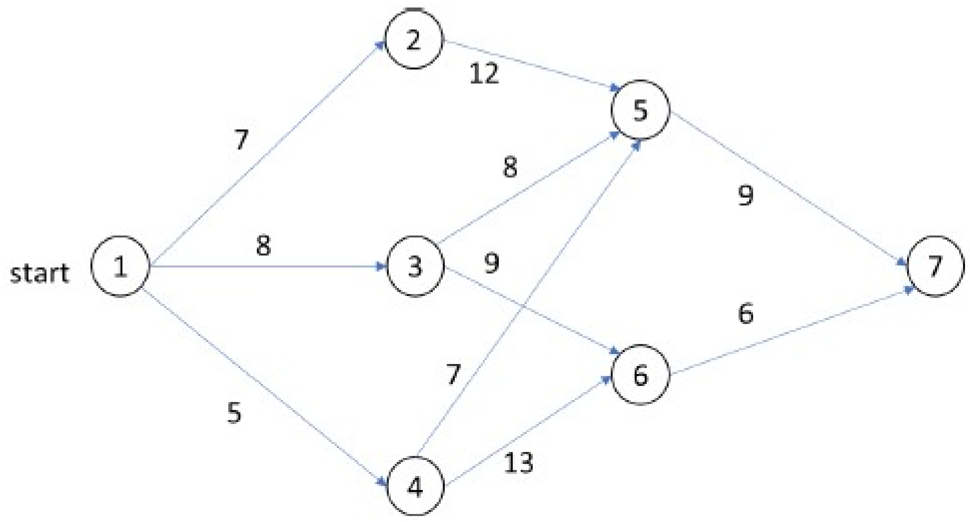

Fig-1

You need to apply dynamic programming (backward recursion) to identify the shortest route. Answer the given subquestions:

Sub questions

Question Number : 23 Question Id : 640653856781 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What does the state variable represent?

Options :

6406532878940. ✖ The city Ramu should visit next in a stage

6406532878941. ✖ Distance covered upto a given stage

6406532878942. ✔ The beginning city in every stages

Question Number : 24 Question Id : 640653856782 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What does the action set represents?

Options :

6406532878943. ✔ The city Ramu should visit next in a stage

6406532878944. ✖ The beginning city in every stages

6406532878945. ✖ Distance covered upto a given stage

Question Number : 25 Question Id : 640653856783 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the objective function value in Stage-2?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
16

Question Number : 26 Question Id : 640653856784 Question Type : MCQ
Correct Marks : 1.5
Question Label : Multiple Choice Question
What is the optimal route?
Options :
6406532878947. ✖ 1 -> 4 -> 6-> 7
6406532878948. ✖ 1 -> 3 -> 5 -> 7
6406532878949. ✔ 1 -> 4 -> 5 -> 7
6406532878950. ✖ 1 -> 3 -> 6-> 7

Sub-Section Number :	5
Sub-Section Id :	640653127964
Question Shuffling Allowed :	Yes

Question Number : 27 Question Id : 640653856785 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Choose the correct term from the Column B for the description given in Column A. Note that one to-one mapping in the table is not necessary

Column A	Column B
a.1 The Walgreen app allows users to manage their pharmacy prescriptions, fill out rapid refill requests, find deals on products in the stores, and make orders that they can pick up at the nearest location.	b.1 Impulse buying
a.2 Buying a candy bar at the checkout aisle of a store because you suddenly crave something sweet after seeing it.	b.2 Multi-channel retailing
a.3 After acquiring a new mobile phone, you considered safeguarding it from scratches by investing in a phone cover. Now, the phone and the phone-cover are---	b.3 Complementary product
	b.4 Omni-channel retailing
	b.5 Substitution product
	b.6 Inventory-dependent demand

Options :

6406532878951. ✔ a.1 : b.4, a.2 : b.1, a.3: b.3

6406532878952. ✖ a.1 : b.5, a.2 : b.2, a.3: b.4

6406532878953. ✖ a.1 : b.3, a.2 : b.4, a.3: b.5

6406532878954. ✖ a.1 : b.2, a.2 : b.1, a.3: b.3

Question Number : 28 Question Id : 640653856786 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Select the irrelevant cost(s) to the inventory management decisions

Options :

6406532878955. ✔ A company bought a machine that broke and could not be returned

6406532878956. ✖ Cost of processing an order of raw materials from the supplier

6406532878957. ✖ Cost of the transportation of an order from supplier site to the manufacturing site.

6406532878958. ✖ Carrying cost for holding the items

Sub-Section Number :